

- (d) **Discrete sizing constraint.** Now assume that only the following inverter input capacitances are available:

10fF , 20fF , 40fF , 80fF , 160fF , 320fF , 640fF , 1280fF , 2560fF .

Choose a practical inverter chain using only these values. Compute the effective fanout of each stage and the total normalized delay. [4 points]

$$C_1 = 0.01\text{pF} = 10\text{fF} \rightarrow \text{uses } 10\text{fF inverter.}$$

$$C_2 = 0.0364\text{pF} = 36.4\text{fF} \rightarrow \text{uses } \overset{20}{\cancel{10}}\text{fF inverter.}$$

$$C_3 = 0.1326\text{pF} = 132.6\text{fF} \rightarrow \text{uses } \overset{80}{\cancel{40}}\text{fF inverter.}$$

$$C_4 = 0.4827\text{pF} = 482.7\text{fF} \rightarrow \text{uses } 320\text{fF inverter.}$$

$$C_5 = 1.7577\text{pF} = 1757.7\text{fF} \rightarrow \text{uses } 1280\text{fF inverter}$$

Please see Pg 12 Notes for calculation of fanout for all stages

*"It depends" - Prof Joshi I assume "Excessive capacitance leads to higher inrush current, size & cost" flying-tech.com/blog/

- (e) **Energy-delay design tradeoff.** Suppose a second design uses one extra inverter stage compared with your answer in part (d), and produces almost the same delay but uses larger total gate capacitance. Explain why a designer might reject the second design even if its delay is similar. capacitor-power-supply-design.

Your answer must discuss both dynamic energy and input loading on the previous logic stage. [4 points]

A designer may reject design 2 due to added cost/area required by the extra inverter. Additionally, an extra gate would add another $(f_i + 1)$ tau to the total normalised delay, slowing the inverter chain. Also, the energy lost in fanout increases both by virtue of adding a gate and the larger total gate capacitances increasing the time needed to fill up each inverter & propagate the signal.

Question 5: Logical Effort and Design Tradeoffs [20 points]

You need to design a logic path that computes a control signal Y and drives a large downstream load. The Boolean function is:

$$Y = \overline{ABCD}.$$

The signal A arrives from a minimum-size inverter whose input capacitance is:

$$C_{in,0} = 10 \text{ fF}.$$

The final output Y must drive:

$$C_L = 1.28 \text{ pF}.$$

Assume the following logical effort model:

$$d_i = g_i f_i + p_i$$

where g_i is the logical effort of gate i , f_i is the electrical fanout of gate i , and p_i is the parasitic delay of gate i .

Use the following logical-effort and parasitic-delay values:

$$\text{Inverter: } g = 1, p = 1.$$

$$n\text{-input NAND: } g = \frac{n+2}{3}, p = n.$$

$$n\text{-input NOR: } g = \frac{2n+1}{3}, p = n.$$

You are considering two implementations.

Implementation 1: Use one 4-input NAND gate to compute

$$Y = \overline{ABCD},$$

followed by a chain of inverters if needed.

Implementation 2: First compute two intermediate NAND outputs:

$$X_1 = \overline{AB}, \quad X_2 = \overline{CD},$$

then combine them using a 2-input NOR gate:

$$Y = \overline{X_1 + X_2}.$$

You may add inverters after Y as output buffers if needed. You may also insert an inverter on an internal signal only if you explain why it is logically required.

5a)

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$$y = \overline{ABCD}$$

$$y = \overline{AB + CD} = \overline{A} + \overline{B} + \overline{C} + \overline{D}$$

if we assume that implementation 2's logic is an unchanging block*, then:

$$\left. \begin{array}{l} y_{2,old} = \overline{X_1 + X_2} \\ y_{2,old} = y_1 \end{array} \right\} y_{2,new} = \overline{y_{2,old}} = \overline{\overline{(X_1 + X_2)}}$$

use a NOT gate (inverter) to correct implem. 2.

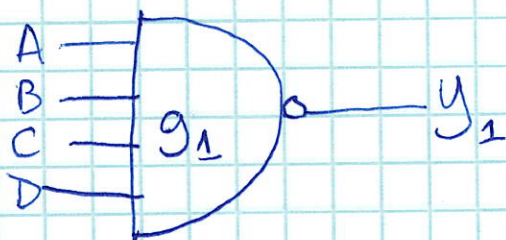
$$\begin{aligned} y_{2,new} &= \overline{\overline{X_1 + X_2}} = \overline{y_{2,old}} & X_1 &= \overline{AB} \\ &= X_1 + X_2 & X_2 &= \overline{CD} \\ &= \overline{AB} + \overline{CD} \\ &= \overline{ABCD} = y_1 \end{aligned}$$

*unless 'logically required'? NOT technically required, can complete without it?

5b

Implementation 1:

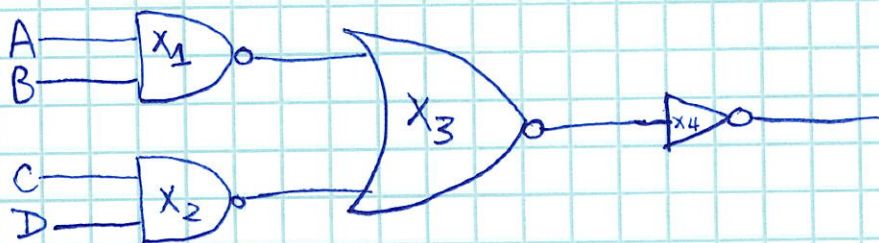
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$N_{\text{input NAND}} : g_1 = \frac{n+2}{3} = \frac{4+2}{3} = 2.$

$G = \prod_i g_i = 2.$

Implementation 2 (corrected)



X_1 is 2 input NAND. $n=2$. $\frac{2+2}{3} = \frac{4}{3}.$

X_2 is 2 input NAND. $n=2$. $\frac{2+2}{3} = \frac{4}{3}.$

X_3 is 2 input NOR. $n=2$, $\frac{2 \times 2 + 1}{3} = \frac{5}{3}.$

X_4 is inverter. $g=1$

$G = \frac{4}{3} \times \frac{4}{3} \times \frac{5}{3} \times 1 = \frac{80}{27}.$

- (a) **Logic correctness.** Verify whether Implementation 1 and Implementation 2 compute the same function. For Implementation 2, explicitly simplify

$$\overline{\overline{AB} + \overline{CD}}$$

and state whether it equals $\overline{ABCD}$. If it does not, propose the smallest correction to Implementation 2 using NAND/NOR/inverter gates. [4 points]

De Morgan!

$$y = \overline{X_1 + X_2} = \overline{\overline{AB} + \overline{CD}} \quad \begin{matrix} X_1 = \overline{AB} \\ X_2 = \overline{CD} \end{matrix}$$

$$\hookrightarrow y = (\overline{X_1}) \cdot (\overline{X_2})$$

$$y = (\overline{\overline{AB}}) \cdot (\overline{\overline{CD}})$$

$$y = (AB) \cdot (CD)$$

See Pg 13 of notes for working out smallest correction to implementation 2

$\hookrightarrow y = ABCD$ not same as $\hookrightarrow y = \overline{ABCD}$
Implementation 2 Implementation 1

- (b) **Path logical effort.** For the corrected versions of the two implementations, identify the path from input A to output Y. Compute the path logical effort

$$G = \prod_i g_i$$

for each implementation before adding output-buffer inverters.

[4 points]

Implementation 1: $y_1 = \overline{ABCD}$ Implementation 2 (New): $y_2 = \overline{\overline{X_1 + X_2}}$

$$X_1 = \overline{AB}$$

$$X_2 = \overline{CD}$$

Please see Notes Pg 14 for working.

$$\text{Imp. 1: } G = \frac{6}{3} = 2.$$

$$\text{Imp 2: } G = \frac{80}{27}$$